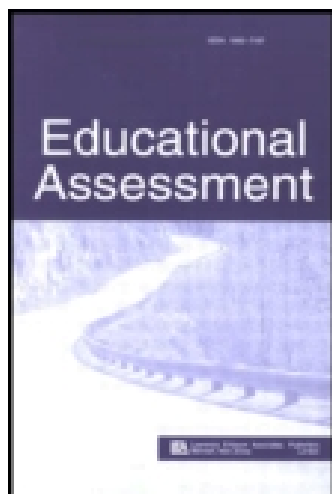




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### Making Sense of Common Test Items That Do Not Get Easier Over Time: Implications for Vertical Scale Designs

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# Making Sense of Common Test Items That Do Not Get Easier Over Time: Implications for Vertical Scale Designs

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This study focuses on an instance in which the mean grade-to-grade scale scores on a vertical scale showed evidence of common test items that do not get easier from one grade to the next. The issue was examined as part of a 2-day workshop in which participants were asked to predict the growth on all linking items used in the construction of vertical scales in math and reading. It was found that these predictions were only weakly correlated to the empirical changes overall. The participants seldom found legitimate rationales for excluding reversal items. Results indicate that a reversal will almost never be sufficient as a rationale for the exclusion of a linking item from the calibration of a vertical scale. Greater attention needs to be given to definition of the construct over which growth is to be interpreted and how this influences the selection of linking items.

Growth is fundamental to education. It is one of the central questions that parents would like a standardized assessment to answer: How much has my child learned over the last year? This question is very difficult to answer in the absence of a test score scale that spans multiple grades. To facilitate such answers, a vertical score scale is typically created after information has been collected about the performance of students in different grade levels on a set of the same items (i.e., a common item nonequivalent group linking design; Kolen & Brennan, 2004). These items, which overlap or are shared across grades, are known as “common items,” or “linking items.” Because inferences about student growth hinge upon the choice of linking items, the criteria used to select them in the design of a vertical scale are very important. It is intuitive to assume that if the same item were to be administered across adjacent grades, the item should become easier to answer correctly by students in the upper grade. But what should be inferred if the item were to become more difficult for students in the higher grade? This phenomenon—a “*p*-value reversal”—occurs whenever the proportion of students answering an item correctly in an upper grade is less than the proportion answering the same item correctly in the lower grade.

When prospective items for a large-scale assessment are being field tested, if a significant number of linking items demonstrate *p*-value reversals, this will have an impact on the resulting trends in grade-to-grade growth. Indeed, given a substantial

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number of linking items with  $p$ -value reversals, mean scale score changes across grades may imply no growth or even negative growth. For example, in a review of growth trends for 16 states with vertically scaled assessments in both mathematics and reading, Dadey and Briggs (2012) found three states with negative or negligible growth across adjacent grades.

The presence of linking items with  $p$ -value reversals brings to the fore a fundamental question about interpretations of growth based on a vertical scale—are negligible or negative patterns of grade-to-grade growth an accurate representation of actual student learning on the construct of measurement, or do they represent a poor choice in the way the scale has been conceptualized and designed? One reaction, which assumes that a poor choice has been made, is to treat a  $p$ -value reversal as a criterion that signals problematic item performance and to exclude items displaying such reversals. A consequence of such a reaction is that the average magnitude of growth will always increase, which could imply that growth is not being measured so much as it is being manufactured. An alternative reaction would be to accept  $p$ -value reversals as an indication that students are not retaining or transferring what they were taught in previous grades, or that they have not learned what was expected in the current grade. In this case, one would accept patterns of no growth or negative growth as an empirical fact that should be, presumably, acted upon.

When  $p$ -value reversals are observed for linking items during a field test, a key question is the extent to which the instruction that students have received is aligned the content of the items. If the items are in fact aligned to the curriculum of instruction, then  $p$ -value reversals may be an indication of test content that is not instructionally sensitive (Polikoff, 2010; Popham, 2007). In this article we demonstrate what we believe to be a relatively novel approach for investigating this issue. We introduce an “item growth workshop” as a means of testing hypotheses about growth and the instructional sensitivity of linking items. In general, we suggest that this can be a productive way for test developers to collect evidence that can be used to validate claims about student growth. More specifically in this article, we use the approach to investigate possible rationales for the inclusion or exclusion of linking items with  $p$ -value reversals.

There are four sections that follow. In the next section we provide some background on different approaches that can be taken in the design of a vertical scale, and we describe the specific nature of two vertical scales in the present study—one in the domain of mathematics and another in the domain of reading. In the second section, we introduce the design of an item growth workshop we convened to investigate the extent to which teachers and curriculum experts are able to differentiate among linking items with and without  $p$ -value reversals. Of particular interest to us was whether the participants would identify a mismatch between the content of an item and the focus of their curriculum and instruction as a plausible explanation for items with  $p$ -value reversals, or, conversely, identify high growth items as those with the closest match to curriculum and instruction. In the third section we present the results from the workshop, which include both the participants’ quantitative predictions about increases in item  $p$ -values across grades and a summary of their qualitative explanations about the performance of items with extreme differences in empirical  $p$ -value changes across grades. In the fourth section we conclude with a discussion of limitations of the current study and recommendations for ongoing research into design considerations for vertical scales.

## CREATING A VERTICAL SCALE

### Overview of the Process

Methods for creating a vertical scale have been well documented (Kolen, 2006; Kolen & Brennan, 2004; Young, 2006). Although there are a number of different ways to create a vertical score scale, the approach generally consists of two interdependent stages: a *design stage* and a *data calibration stage*. In the design stage, two critical decisions must be made. First, a definition needs to be established for growth—what is it that is growing exactly? Kolen and Brennan (2004) distinguished between two different ways that growth could be conceptualized in the design stage. They introduced what they called the “domain” and “grade to grade” definitions of growth. In what they refer to as the domain definition of growth, the term domain encompasses all of the content covered across some collection of grade level content standards. Thus growth is defined over the full content of the domain. Operationally speaking, one implementation of the domain definition would require that students in each grade level take a sample of items targeted to the standards on each and every grade. In contrast, Kolen and Brennan introduced the grade-to-grade definition of growth for the scenario in which growth is defined only on the subset of the full content domain covered by the standards that overlap in adjacent grades. For example, although the full domain of mathematics may include subdomains of content related to number sense, measurement, and geometry, growth from Grades 3 to 4 could be defined solely from items aligned to standards for number sense and measurement, whereas growth from Grades 4 to 5 could be defined solely from items aligned to standards for measurement and geometry.

Briggs (2011) suggested a “learning progression” definition of growth that has some similarity to Kolen and Brennan’s domain definition in the sense that a learning progression design focuses upon growth with respect to a common definition of focal content across grades. However, the learning progression approach departs from Kolen and Brennan’s domain definition in that it emphasizes one “big picture” concept (or collection of related concepts) at a time and a cognitive model for how students become more sophisticated in their understanding and applications of this concept as they are exposed to instruction. The domain definition generally lacks this type of specificity, as it attempts to characterize growth across a large content area, such as “mathematics proficiency as operationalized by content standards.” In addition, the implementation of the domain definition generally does not articulate a cognitive model for how students become more sophisticated in their understanding and application of any given concept.

The second critical decision in the design stage deals with data collection. Although there are three data collection designs applicable to vertical scaling (common item, equivalent groups, and scaling test designs; Kolen & Brennan, 2004), we focus on the common item design, which has been used by at least 16 states with vertically scaled, statewide assessments (Dadey & Briggs, 2012). Given the use of a common item design, the second critical decision deals with the selection of the set of linking items that will be administered to students across two or more adjacent grade levels. Linking items stand in contrast to unique items, which are administered only to students in any single grade. In some designs, linking items consist of an external test (i.e., a scaling test) given to students across multiple grades, in others they consist of an external test given only across adjacent grades, and in others they consist of items embedded within operational test forms. When items span adjacent grades, an important decision is whether the linking items overlap across grades in a “backward” direction, a “forward” direction, or both.

When linking items overlap in a backward direction, students in an upper grade are taking items with content most directly targeted to instruction from the lower grade. This is illustrated in Figure 1. In the Figure (and in Figure 2), hypothetical linking items for Grades 4 and 5 are used as an example. The “O” stands for the outcome that would be observed when a student answers a set of linking items, and this outcome could be expressed for each student as the proportion of items answered correctly. The subscript 4 indicates that the outcome is based on items written to correspond to Grade 4 content. This is called a “backward” linking design, because Grade 5 students take items that were written for Grade 4 students. The backward linking approach makes it possible to answer one specific question that is usually not answerable in a testing context: “How would the average Grade 5 student perform if they were tested on Grade 4 content?” When compared to the performance of current Grade 4 students on the same items, it becomes possible to make an inference about growth on what amounts to a delayed posttest experimental design: Students take a posttest following Grade 4 instruction, and this is contrasted to the same posttest for a different group of Grade 5 students after Grade 5 instruction. Notice that in this design there is no pretest being given on Grade 4 content *before* a student has been exposed to Grade 4 instruction. Figure 1 includes the experimental design implied by this

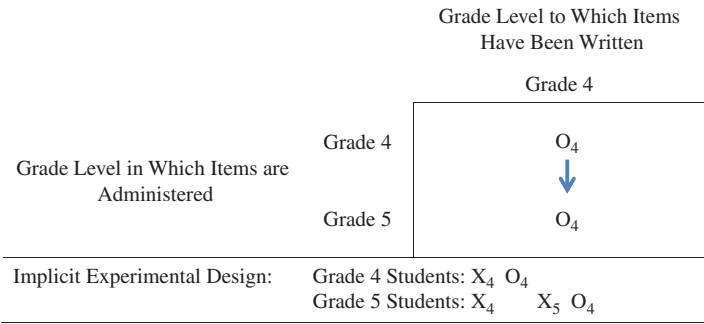


FIGURE 1 A “backward” item linking design and the implied experiment.

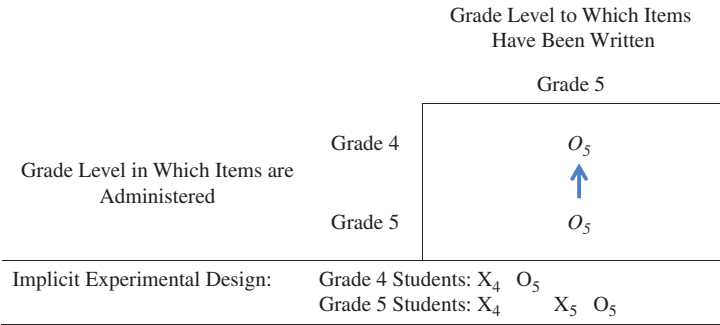


FIGURE 2 A “forward” item linking design and the implied experiment.

linking structure, where an “X” indicates exposure to grade-specific instruction and neither group has been randomly assigned.

When linking items overlap in a forward direction, students in a lower grade are taking items with content targeted to the curriculum and instruction students are anticipated to receive in the upper grade. As illustrated in [Figure 2](#), this is called a “forward” linking design because Grade 4 students take items that were written for Grade 5 students. The forward linking approach makes it possible answer a different question that is usually not answerable in a testing context: “How would the average Grade 4 student perform if they were tested on Grade 5 content?” In this sense, the scores of Grade 4 students amounts to a pretest that occurs prior to Grade 5 instruction. When compared to the performance of current Grade 5 students on the same items after a year of instruction, it becomes possible to make an inference about growth in Grade 5 using what amounts to pre-post experimental design.

Once item responses have been gathered for students who are representative of each grade level, the next task is to analyze differences in performance on the linking items. These differences become the basis for the data calibration stage. To calibrate the responses from students at different grade levels onto a single scale, either the ability of the students or the characteristics of the items (e.g., difficulty) needs to be held constant across grades. Because growth in student ability across successive grades is the underlying basis for the vertical scale, the only reasonable candidate to be held fixed are item characteristics. Item response theory (IRT)-based methods (Lord & Novick, 1968) are by far the predominant approach and have been since the mid-1980s. A desirable aspect of IRT is the property of parameter invariance, which will hold so long as the assumption of local independence has been satisfied, and the data can be shown to fit the item response function that has been specified. Parameter invariance is the critical property of IRT models that makes it possible to establish values for the characteristics of common items that do not depend on the particular group of students responding to them. When parameter invariance holds, the same difficulty parameter (up to a linear transformation) will be estimated for an item whether it is administered to a third-grade student or an eighth-grade student.

Much of the research literature on vertical scales has focused on choices that must be made in the calibration of the scale (see Skaggs & Lissitz, 1986, for a review along these lines). Two choices in particular have received considerable attention: the functional form of the IRT model, and the manner in which tests scores across grades are concatenated. The first choice is typically a contrast between the use of the three parameter logistic model (Birnbaum, 1968) or the Rasch Model (Rasch, 1960). The second choice is a contrast between a separate or concurrent calibration approach. In the separate approach, item parameters are estimated separately for each grade-specific test. Then a base grade for the scale is established, and other grades are linked to the base grade after estimating linking constants for each grade-pair, almost always by using the Stocking-Lord approach (Stocking & Lord, 1983). In the concurrent approach, all item parameters are estimated simultaneously. There are also hybrid permutations of these approaches. For example, Briggs and Weeks (2009) did concurrent calibration for the same grade across years and then did separate calibration for links across the concurrently calibrated grades. Another example would be calibrating a scale after importing fixed item parameters from a previously calibrated vertical scale. There is very little consensus in the research literature about the best way to calibrate a vertical scale, and different permutations of approaches, applied to the same data, have been shown to impact the magnitudes of grade-to-grade growth inferences (Briggs & Weeks, 2009).

Most previous research on vertical scales has focused on choices that psychometricians must make during the data calibration stage: exploring the sensitivity of the scale to IRT modeling decisions or to violations of IRT assumptions and properties such as local independence, unidimensionality, and parameter invariance (Briggs, 2013; Camilli, 1988, 1999; Camilli, Yamamoto, & Wang, 1993; Yen, 1985; 1986; Yen & Burket, 1997). Somewhat surprisingly, research that attempts to evaluate the design stage and propose ways that this could be improved appears nonexistent, nor could we find any published studies that discuss the issue of linking item reversals. The present study represents a first step in filling this void in the research literature.

### Design of the Field Test in the Current Study

*Student sample participating in field test.* The present investigation focuses on actual vertical scales created for one state's criterion-referenced reading and mathematics assessments across Grades 2 to 10. The state in question is relatively small, with a population of roughly 1 million residents, 100,000 students, and 8,000 teachers in 200 public schools. A computer-administered field test was conducted in 2010 to gather data that could be used to calibrate vertical scales and establish a pre-equated item bank. According to the state's test contractor, "All students in the state were eligible to participate in the field test. . . . The aim was to achieve at least a 75% participation rate among all eligible students in all grades, although this target was surpassed." Although this is generally true, in an independent evaluation of the field test sample conducted prior to the study we present here, it was found that the proportion of the eligible population of students who choose to participate decreased significantly by grade level, from above 90% in Grades 2 through 5 to about 84% in Grades 6 to 8, and finally to about 77% in reading and 70% in math by Grades 9 and 10.

Unfortunately, data were not available for us to compare the demographic and prior academic achievement profiles for student/schools who did and did not participate in the field test. If higher achieving students were less likely to participate in upper grades, this would bias the growth implied by a vertical scale downward and could help to explain the presence of linking items with  $p$ -value reversals. Conversely, if lower achieving students were less likely to participate, this would bias growth upward and potentially mask the presence of additional linking items with  $p$ -value reversals. However, the difference in ability between participants and nonparticipants would have to be rather strong before it would be likely to have a significant impact. For example, given a mean reading growth across Grades 9 to 10 of 0.10 logits for the participating sample, if this growth would have been twice as large—0.20 logits—for the nonparticipating sample, than the true growth would still only be  $.77 \times .1 + .23 \times .2 = .12$  logits.

*Stratified sampling of items.* Thousands of items per subject were field tested using a matrix sampling design and stratified random item selection for "on the fly" test form construction. Students in Grades 3 and higher took a field test consisting of 35 items, whereas students in Grade 2 took only 20 field-test items. After stratifying on the gender and race/ethnicity of participating students, items were randomly selected from the item pool using an algorithm with constraints imposed to ensure that each item was not over- or underexposed. The process was summarized by the test contractor as follows:



Upon test startup, the algorithm selects a series of items in the following iterative sequence:

- 1. Identify all the items that were least frequently administered to the stratum.
- 2. Randomly select an item from among that group with equal probability.
- 3. Return to Step 1 and continue if the requisite number of items has not been met.

Subsets of the total number of items, about 500 in total per subject, were administered in adjacent grades as linking items. The full sample of students per field test grade was between 7,000 and 9,000, and each linking item was taken by anywhere from 500 to 1,500 students. The lower bound of 500 on sample size per item was considered adequate because the state was planning to calibrate the items using the Rasch Model.

*P-value reversals.* When the properties of the linking items were examined after the field test, the amount of growth that the items implied was, in general, quite small. It was relatively rare to find examples of items for which the percentage of students answering the item correctly increased by 10 or more percentage points, and about 20% of linking items in reading and 10% in math had *p*-value reversals. This was true even after excluding linking items because they demonstrated poor fit to the Rasch Model, or because problems were found with the wording of the items as part of a content review (these and other criteria used to scrutinize the items are described in more detail in a subsequent section).

The decision of whether to include or exclude items with *p*-value reversals has practical consequences for the interpretation of grade-to-grade growth trends in this state, particularly in reading, where there were twice as many linking items with reversals as there were in math (100 linking items in reading and 53 in math had reversals, out of 496 and 480 linking items, respectively). Figure 3 plots two different reading growth trends across Grades 2 to 10. The units on the vertical axis represent grade-to-grade gains as a proportion of the pooled standard deviation across grades (i.e., grade-to-grade effect-sizes). The solid line shows the

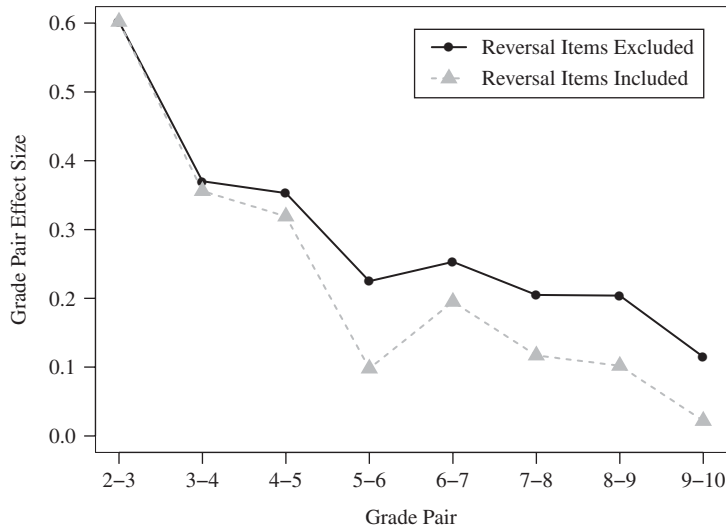


FIGURE 3 Reading effect-size growth trends by *p*-value inclusion.

resulting growth trend if linking items with reversals were removed prior to calibrating the vertical scale; the dashed line shows the trend if they are kept. Without the reversal items, the gains from Grade 6 on tend to be about 0.10 *SDs* higher on average than they would be otherwise.

## Research Questions

In the following sections we report on the results of a workshop that we conducted with a purposeful sample of the state's Grade 2 to 10 teachers and curriculum experts (whom we refer to henceforth as "participants.") The purpose of this workshop was to ask participants to predict growth at the item level, in terms of percentages correct, and then to examine items with actual *p*-value reversals or high growth in light of these predictions. Another goal was to develop content-based rationales for the inclusion or exclusion of items with *p*-value reversals. We posed three specific research questions:

1. How well does the actual growth observed for linking items on the field test compare to the growth predicted by participants?
2. To what extent can participants distinguish items with and without *p*-value reversals, before seeing impact data?
3. To what extent can participants provide plausible content-based explanations for *p*-value reversals?

The idea behind the first research question is similar in spirit to the question posed in most standard setting panels when impact data are provided after initial cutpoints have been set. It is often the case that panelists are surprised to find that the level of student performance in any given grade is lower than was anticipated. Our results are analogous to these results in the context of growth: Participants tend to predict significantly more growth from grade to grade than is actually observed. The idea behind the second question is to see if items with *p*-value reversals stand out to participants when they are mixed together with the full pool of linking items. If they do not stand out, this is evidence against the blanket exclusion of items with *p*-value reversals. In fact, we find that participants do not distinguish reversal items from the full pool of items in either subject, although they are able to distinguish low growth items from high growth items in math. Finally, with respect to our third question, participants were not able to provide content-based explanations that would suggest items with reversals were systematically misaligned to the curriculum.

## METHODS

### Reading and Math Field Test

The vertical scales we examined spanned Grades 2 through 10 in mathematics and reading. The initial pool of candidate linking items was sampled from the total pool of available items, resulting in an initial total of 488 and 480 candidate linking items in reading and math, respectively. The candidate pools of linking items appear to have been sampled from the content of each grade, making it somewhat unclear whether growth should be given a grade-to-grade or domain-level interpretation. In the full linking item pool, there are approximately 60 common items per grade (excluding Grades 2 and 10, where the number of common items is about 30).

In support of this study we were given access to the linking items themselves (i.e., both the prompts and the responses), as well as a good deal of other information relevant to item development and screening process. This included information about whether items were forward or backward linking (the items were evenly divided in this manner). In addition, we were given item flags that were grouped by the test contractor into four categories. The first flag indicated whether item or passage *content* material was inappropriate and was generated by the state department as part of their item review. The second flag, *data review*, is based primarily on classical item statistics. Because the linking items have two sets of statistics (one from the upper grade and one from the lower grade), a linking item is flagged if one of the following issues occur in any of the two possible grades: (a) point-biserial for correct response less than 0.20, (b) point-biserial for any distractor greater than 0.05, (c)  $p < .25$  or  $> .95$ , (d) evidence of differential item functioning by gender, and (e) any distractor percentage is greater than the percentage for the correct answer. The third flag indicates *misfit* to the Rasch model (infit or outfit statistics less than 0.8 and greater than 1.2) used to calibrate the scale in a concurrent calibration across all grades. The fourth and final flag, *p-value reversal*, is the focus of our investigation.

Each item was identified as addressing a single content standard. The content standards were subdivided into performance indicators (PI). The PIs were subdivided further into grade-level expectations, which describe what students should know and be able to do in somewhat concrete terms. In an analysis not presented here we found that the candidate linking item pool was generally representative of the PIs of the total item pool, which is as would be expected given the linear on the fly test design used for field test forms.

### Item Growth Workshop Design

Workshop participants in reading and math were recruited according to the following stipulations: (a) a minimum of 5 years' experience teaching, (b) at least four participants per grade grouping or band,<sup>1</sup> and (c) participants should come from different schools and school districts across the state. In reading there were 25 workshop participants (17 teachers, eight curriculum specialists) from 11 unique school districts. In math there were 23 workshop participants (nine teachers, 14 curriculum specialists<sup>2</sup>) from 14 unique school districts. Out of the 48 participants, 92% were female and 12% were non-White. Among the active teachers, the median number of years of experience was 17, and the minimum was 5.

The workshop began with an overview of the activities that would be taking place over the course of the 2 days. Participants were told that the purpose of the workshop was to gather information to evaluate the validity of grade-to-grade growth interpretations along the state's reading and mathematics scales. We then explained that on the 1st day of the workshop participants would be asked to work independently on the items to predict the changes in student performance they would expect to see across adjacent grades. Participants were informed that on the 2nd day they would be given the opportunity to compare their predictions of item growth with the growth that was actually observed on the field test. As part of this orientation session, participants were given a conceptual overview that explained the use of

<sup>1</sup>A grade band is a set of two to three adjacent grades.

<sup>2</sup>Because of difficulties in recruiting a sufficient number of teachers in math, a larger proportion of district-level curriculum specialists were recruited to participate than in reading.

linking items and their role in forming a vertical scale. The session concluded with an opportunity for participants to practice making item-level growth predictions (described in more detail next), and then ask clarifying questions. On the 1st day of the workshop, participants were intentionally not given any information about the actual distribution of  $p$ -value changes on field-tested common items, nor did we reveal that a significant number of these items had resulted in  $p$ -value reversals. The only empirical information given to participants at the outset was descriptive statistics for the full distribution of  $p$ -values for grade-specific items in reading and math.

Following the orientation session, participants were separated into small groups by subject and grade band. There were four grade band groups: 2–4, 4–6, 6–8, and 8–10. Within a group, participants were given hard copy booklets containing linking items for adjacent grades. For example, the 4–6 group was given two booklets of linking items, one booklet for items common to Grades 4 and 5 and another with items common to Grades 5 and 6. Along with these item booklets, participants were given a separate scoring sheet, with each row of the scoring sheet associated with a distinct linking item in the booklet.

Figure 4 illustrates the format of a given row on the scoring sheet. For each item, participants were given the wording of the actual item, information about the associated content standard underlying the item, and the observed  $p$ -value, as a percentage, in the lower grade. In addition to the columns above, the rating sheet also contained a column for notes. Participants were asked to follow the same general thought process in making a prediction for the upper grade  $p$ -value:

Imagine that the same students are given this item near the end of Grade 4 and again near the end of Grade 5. Assume that the instruction the students receive in Grade 5 is ‘typical’ for a Grade 5 classroom in the state.

It turns out that 76% of these students answer the item correctly at the end of Grade 4. What percent of these same students would you expect to answer this item correctly by the end of Grade 5?

The preceding example corresponds to a forward linking design (recall Figure 2), and we used this throughout as the prompt to get participants to make upper grade  $p$ -value predictions. Limiting the scenario to a forward linking design was a possible weakness in our design; at the time it was a sacrifice we thought was necessary to keep the task for participants clear. Had we instead given them a scenario corresponding to the design for backward linking items for those items that were, in fact, backward linking items, it is quite possible that this could have changed

| ID | Passage              | Standard | PI                                                               | GLE                                                        | Actual Grade 4 Performance (% Correct) | Predicted Grade 5 Performance (% Correct)? |
|----|----------------------|----------|------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------|--------------------------------------------|
|    | A Tracking Adventure |          | Figurative Language and Non-literal Expressions (Interpretation) | : Differentiating between literal and non-literal meaning. | 76                                     |                                            |

FIGURE 4 A row in the item prediction sheet given to groups of 4–6 participants in reading. *Note.* The math sheet omitted passage information. Some information has been blacked out to conceal the state’s identity.

their predictions. The trade-off is that this could have also invited confusion as it would have become clear that each type of linking item was posing a different “experiment” as basis for making inferences about growth. We revisit this issue in the final section of the article.

Participants were explicitly asked to complete the item growth predictions on the 1st day independently, and each group had a “table leader” to monitor the process and thereby ensure that there was no collaboration among participants during this first stage. Participants were given a blank space to the right of each item to write down any observations about the quality of the item. The breakdown of raters by adjacent grade common item is given in Table 1.

Participants had up to 5 hr to make growth predictions for all candidate linking items that were administered in the field test. This turned out to be ample time to complete the task.

On the 2nd day of the workshop, the small groups were provided with a spreadsheet for each of the two adjacent grade common items they had rated. These spreadsheets contained (a) the actual lower and upper grade  $p$ -values for the 10 linking items with the least and most observed growth, 20 items in all, and (b) the corresponding  $p$ -value predictions they had made the day before. Table leaders were instructed to facilitate and take notes on a discussion based on the following motivating questions:

- Irrespective of your initial predictions, can your group speculate about explanations rooted in curriculum and instruction for the fact that students appear to show significant growth on some items but negative or negligible growth on others?
- Do you think that the items with negative growth should be excluded? Why or why not?
- How much time do you spend teaching the content represented by the high- and low-growth items, respectively?

Each group was given 4.5 hr to engage in these discussions. Participants then reassembled by subject area, and each group was given the opportunity to summarize the results from their discussions and comment more generally on the 2-day process.

We excluded a small subset of 41 items from the initial candidate set in math because these were constructed response items that the state had decided to exclude from consideration as a basis for calibrating the vertical scale prior to content or data review. The remaining 488 and 439

TABLE 1  
Number of Raters by Common Item Group

| <i>Common Item Pair</i> | <i>Reading</i>      |                 | <i>Math</i>         |                 |
|-------------------------|---------------------|-----------------|---------------------|-----------------|
|                         | <i>No. of Items</i> | <i>N Raters</i> | <i>No. of Items</i> | <i>N Raters</i> |
| 2–3                     | 58                  | 6               | 60                  | 6               |
| 3–4                     | 64                  | 6               | 60                  | 6               |
| 4–5                     | 60                  | 5               | 60                  | 5               |
| 5–6                     | 65                  | 8               | 60                  | 5               |
| 6–7                     | 64                  | 8               | 60                  | 6               |
| 7–8                     | 62                  | 5               | 60                  | 6               |
| 8–9                     | 63                  | 6               | 60                  | 6               |
| 9–10                    | 52                  | 6               | 60                  | 6               |
| Total unique            | 488                 | 25              | 480                 | 23              |

reading and math linking items remain constant throughout, with all items in a multiple-choice response format.

RESULTS

The boxplots in Figures 5 and 6 summarize the empirical distributions of actual growth on all forward and backward linking items that were administered in the field test by grade pair for reading and math.<sup>3</sup> As noted earlier, the changes in *p*-values across grades tend to be very small in an absolute sense. Only from Grades 2 through 4 do we observe a median change that is above 5 percentage points in both reading and math. For each grade pair, an equal mix of roughly 30 forward and 30 backward linking items were administered, and *p*-value change distributions are disaggregated accordingly. In math, where there is more variability in *p*-value changes than in reading, forward linking items generally have higher *p*-value increases than backward linking items, though the two distributions have considerable overlap in all grade pairs except from 3 to 4 and from 6 to 7. In reading, the pattern is somewhat similar, but from Grades 3 to 4 *p*-value changes on backward linking items are significantly higher than those for forward linking items. One notable finding is that from Grades 7 to 10, *p*-value reversals are far more likely on backward linking items. In math, where *p*-value reversals were on the whole less common, the interaction between reversal and linking design was not as evident.

Figure 7 compares, by subject, the distributions for the actual common item *p*-value changes across all grades relative to the distributions of the mean *p*-value changes that were predicted by

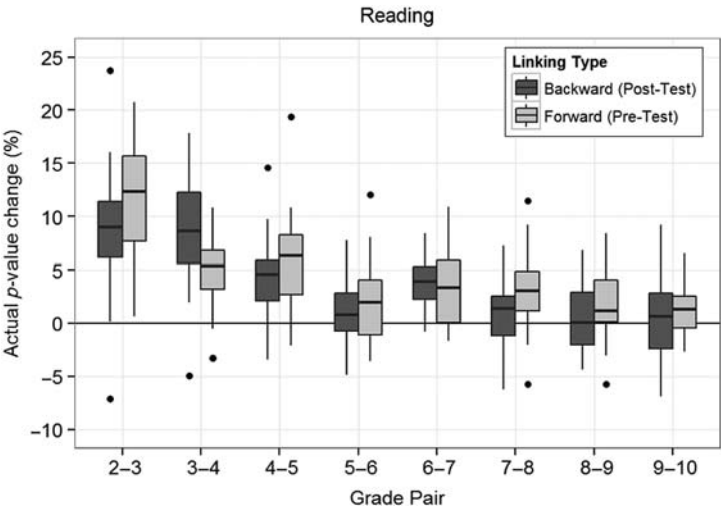


FIGURE 5 Distributions of reading linking item growth by type and grade.

<sup>3</sup> There is a small exception to this statement: In math there were three linking items that crossed three grades so that they were both forward and backward linking. These three items were excluded from Figure 6.

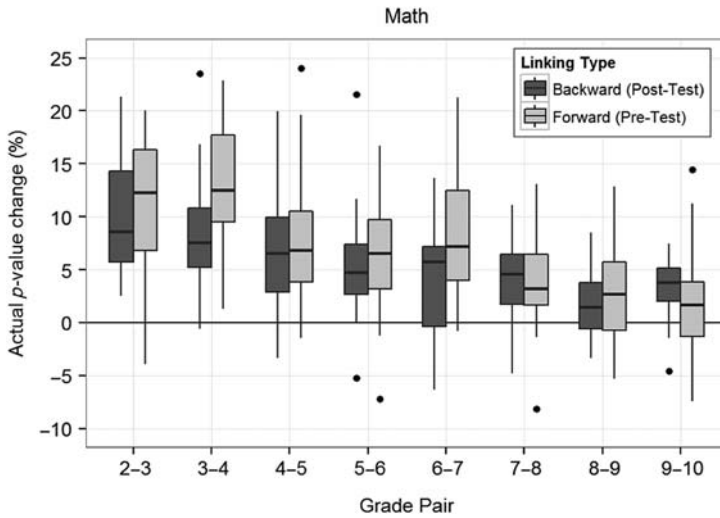


FIGURE 6 Distributions of math common item growth by grade.

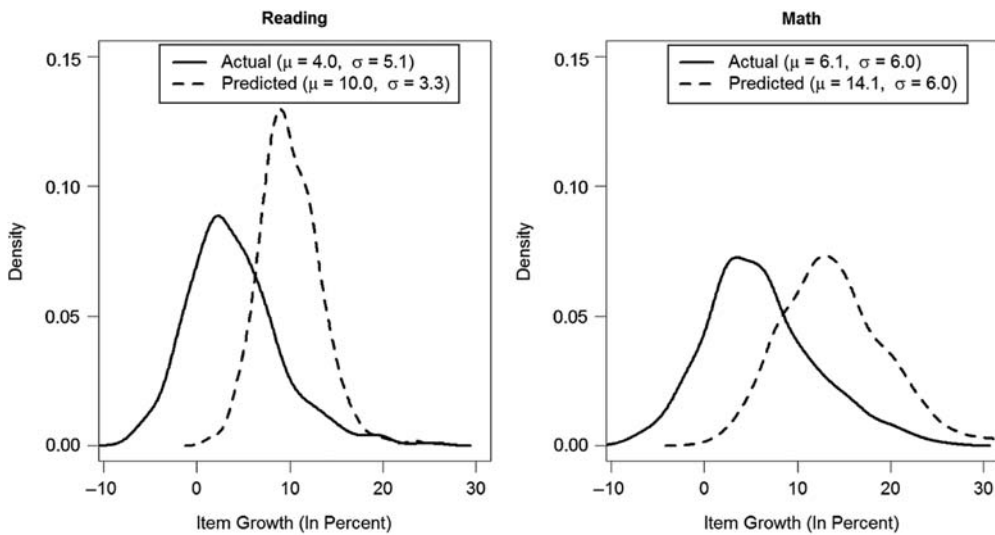


FIGURE 7 Comparison of distributions of predicted and actual item growth. *Note.* Linking items are units of analysis.  $N = 488$  for reading,  $N = 439$  for math.

each group of raters. At virtually all points along the empirical distribution, participants in the workshop predicted significantly more growth on common items than was actually observed. On the other hand, the variability of mean predictions was quite similar to the empirical variability in observed growth (indeed, for math the values are identical).



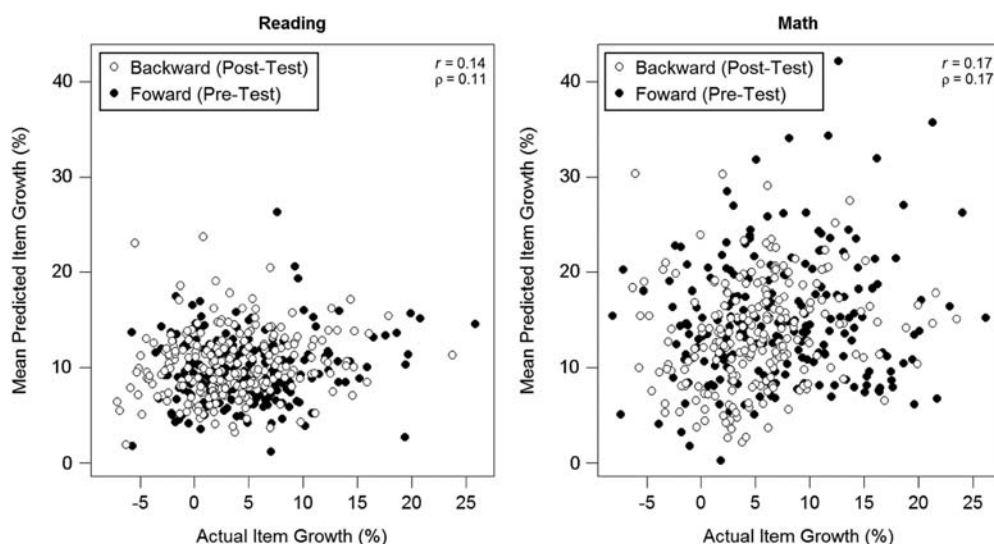


FIGURE 8 Scatterplots of mean predicted item growth and actual item growth.

On the whole, workshop participants were not able to reliably distinguish linking items with respect to their growth. The correlation of observed and predicted  $p$ -value growth was weak for both subjects, as evident in the scatterplots shown in Figure 8. In reading, Pearson and Spearman correlations were 0.14 and 0.11; in math, both Pearson and Spearman correlations were 0.17. Although small, all correlations were statistically significant with  $p < .05$ .

One other interesting comparison is in the relationship between the lower grade  $p$ -value and the  $p$ -value change from the lower to upper grade. In reading, the correlation between lower grade  $p$ -values and teacher growth predictions ( $r = -.16$ ) matches the actual correlation between lower grade  $p$ -values and actual growth almost exactly ( $r = -.21$ ). By contrast, math teachers were more strongly influenced by the lower grade  $p$ -value in their predictions of growth: The correlation between lower grade  $p$ -values and predicted growth was  $-0.40$ , but the correlation with actual growth was just  $-0.10$ . In other words, math teachers were much more likely to predict lower growth for items they felt were “too easy.”

To better understand the factors that led teachers to predict high growth for an item, we conducted a regression analysis in which linking items by test subject were the units of analysis, and mean *predicted* growth for each item (in percentage) was always the outcome variable of interest. The key predictor of interest in each regression was a dummy variable that indicated whether a linking item actually showed a  $p$ -value reversal across adjacent grades (“reversal item”). We controlled for the following factors: a covariate for lower grade  $p$  value; a dummy variable for whether the item was forward linking; dummy variables for grade pair (with the Grade 2–3 pair as the omitted category); and dummy variables for items flagged for problems with content, item statistics, or misfit to the Rasch Model. If, holding these factors constant, workshop participants can discriminate reversal items from all nonreversal items we would expect the parameter estimate for the reversal variable to be negative and statistically significant.



TABLE 2  
Factors Explaining Variability in  $p$ -Value Change Predictions: All Items

| <i>Variable</i>       | <i>Reading</i> | <i>Math</i>   |
|-----------------------|----------------|---------------|
| Constant              | 13.77* (0.66)  | 17.70* (1.30) |
| Reversal item         | -0.33 (0.35)   | -0.93 (0.67)  |
| Base $p$ (in %)       | -0.04* (0.01)  | -0.14* (0.01) |
| Content               | 1.42* (0.42)   | -4.75* (0.90) |
| Misfit to Rasch model | -1.41* (0.41)  | -0.85 (0.81)  |
| Data review           | -0.51 (0.40)   | 0.30 (0.59)   |
| Forward linking       | -0.45 (0.27)   | 0.32 (0.46)   |
| Grade pair 3-4        | -1.03 (0.54)   | 1.85* (0.91)  |
| Grade pair 4-5        | -3.92* (0.54)  | 4.52* (0.92)  |
| Grade pair 5-6        | -1.95* (0.54)  | 5.59* (0.93)  |
| Grade pair 6-7        | -1.37* (0.54)  | 9.31* (0.93)  |
| Grade pair 7-8        | 0.49 (0.54)    | 6.08* (0.93)  |
| Grade pair 8-9        | 0.37 (0.55)    | 2.24* (0.99)  |
| Grade pair 9-10       | -1.74* (0.57)  | 1.75 (0.96)   |
| $R^2$                 | 0.27           | 0.44          |

*Note.*  $N = 488$  items for Reading, 439 items for Math. The outcome variable is mean growth predicted by workshop participants ( $SD = 3.3$  in Reading,  $SD = 6.0$  in Math).

\* $p < .05$ .

As can be seen in the results shown in Table 2, although the coefficient for the reversal variable is negative for both reading and math, it is not statistically significant. In other words, workshop participants did not draw significant distinctions between reversal items and nonreversal items in their predictions of linking item growth. In contrast, in both reading and math, participants did make significant distinctions between linking items that had previously been flagged for problems with content (though curiously, in reading participants actually predicted *higher* growth for such items) or linking items that were above or below average in difficulty ("Base  $p$ "). In reading, participants also predicted significantly lower growth for items that were ultimately flagged as misfitting the Rasch Model.

It can be reasonably argued that expecting participants to discriminate reversal items from all nonreversal items is highly unrealistic because the actual distinction on either side of a  $p$ -value difference of zero is relatively arbitrary. This is a consequence of sampling error. To think about this, if one uses as a lower bound a sample size of 500 students taking the same item in a lower and upper grade, then depending upon the starting and ending  $p$ -value percentages, the standard error on a  $p$ -value difference will be about 2 to 3 percentage points. For a larger sample (some items had about 1,500 respondents), the standard error would be about 1.5 percentage points. So to be conservative, one might argue that two items should differ in their actual  $p$ -value changes by at least 5 percentage points before one should be considered a true "low-growth" linking item and the other a true "high-growth" item. This can be accomplished by restricting, for each grade pair, the linking items under consideration to those in the bottom and top quartiles of growth. To what extent do workshop participants discriminate low-growth items (which include both reversal and nonreversal items) from high-growth items?

The answer is that workshop participants appear to have been able to discriminate between low- and high-growth items in math but not in reading. On average, as shown in Table 3, math

TABLE 3  
Factors Explaining Variability in *p*-Value Change Predictions: Top and Bottom Quartiles of Growth

| <i>Variable</i>       | <i>Reading</i> | <i>Math</i>    |
|-----------------------|----------------|----------------|
| Constant              | 13.89 (1.06)   | 16.66 (1.94)   |
| Low growth item       | − 0.47 (0.41)  | − 2.73* (0.76) |
| Base <i>p</i> (in %)  | − 0.04* (0.02) | − 0.12* (0.02) |
| Content               | 1.93* (0.69)   | − 3.36 (1.72)  |
| Misfit to Rasch model | − 1.35* (0.68) | 0.17 (1.08)    |
| Data review           | − 0.24 (0.64)  | 0.28 (0.83)    |
| Forward linking       | − 0.72 (0.42)  | 0.92 (0.68)    |
| Grade pair 3–4        | − 1.39 (0.82)  | 2.32* (1.34)   |
| Grade pair 4–5        | − 3.51* (0.81) | 5.29* (1.40)   |
| Grade pair 5–6        | − 1.25 (0.79)  | 5.96* (1.42)   |
| Grade pair 6–7        | − 0.82 (0.81)  | 10.86* (1.46)  |
| Grade pair 7–8        | 0.55 (0.80)    | 6.65* (1.38)   |
| Grade pair 8–9        | 0.53 (0.80)    | 4.17* (1.44)   |
| Grade pair 9–10       | − 1.18 (0.85)  | 3.01* (1.43)   |
| <i>R</i> <sup>2</sup> | 0.24           | 0.50           |

*Note.* *N* = 246 items for Reading, 222 items for Math. The outcome variable is mean growth predicted by workshop participants (*SD* = 3.3 in Reading, *SD* = 6.0 in Math). Values in parentheses are standard errors.

\**p* < .05.

participants predicted *p*-value changes for top quartile linking items that were 3 percentage points higher than the changes they predicted for bottom quartile linking items. This difference is both statistically and practically significant. Among reading participants, the difference in predicted *p*-value changes for top and bottom quartile items was neither statistically nor practically significant. These results suggest that it would be hard to make a case for the exclusion of reversal items in reading on the grounds that they are perceived by teachers and content experts as lacking instructional sensitivity. Workshop participants did not perceive a difference between low- and high-growth items when presented with them blindly and asked to predict which would demonstrate more growth. In math, participants did perceive a difference. Next we turn to the explanations that each group of participants gave for differences between low- and high-growth items when they were presented with actual results from the field test.

### What Explanations Do Participants Provide for Linking Items With Low or No Growth?

**Reading.** Out of the 28 participants who rated common item growth in reading, 95% indicated that they were “surprised” or “very surprised” by the difference between the item growth they had predicted and the item growth that was observed. One participant remarked that it was very hard, in the context of predicting growth, to separate “what kids *do* know from what kids *should* know.”

When asked to explain why some items had *p*-value reversals, two explanations were most commonly invoked: student motivation and overanalyzing. Participants on the whole believed that students’ motivation to give their best effort was likely to decrease as grade level increased.

In particular, many participants commented that the reading passages on the field test tended to become lengthier and less likely to engage students in the upper grades. Some items with reversals were ones that required students to scroll back to a particular section of the passage to find the information needed to answer the question, and participants suspected that middle school students would be more likely to guess at an answer rather than take the time to double-check what was in the passage relative to elementary school students. The propensity for “lazy reading” combined with increasingly complex and lengthy reading passages was viewed as a bad combination as students enter middle school.

Participants also proposed that a good number of items (both with and without reversals) involved terminology that was inconsistent with the language used instructionally by teachers. For example, many items with reversals used the terms “most likely,” “probably,” or “mainly” when asking students to make inferences or draw conclusions from a reading passage. Participants claimed that these were terms that were inconsistent with the way students are taught to comprehend a reading passage and were likely to trigger associations with mathematical reasoning. This could have led students in later grades to overanalyze some items, assuming the correct answer was too obvious to be correct.

Given the empirical results presented earlier, which indicate that participants were unable to distinguish low- and high-growth items when presented with them blindly, it is entirely possible that participants were retrofitting a cause (decreasing motivation, overanalyzing) for a known effect (negligible growth). Indeed, for a majority of items with reversals, participants were at a loss to come up with a plausible explanation. For example, on a Grade 6 to 7 common item for which the  $p$ -value dropped from 76% to 74%, the consensus expressed was “This is really easy. We cover this [material] in 7th grade. There is no real reason for the decrease.”

**Mathematics.** Out of the 23 participants who rated common item growth in reading, 80% indicated that they were “surprised” or “very surprised” by the difference between the item growth they had predicted and the item growth that was observed. There was a degree of overlap between the mathematics teachers’ explanations of why some items had  $p$ -value reversals and the explanations provided by reading teachers. Namely, for some items the teachers either had no explanation or offered an explanation that would not explain the reversal because it would apply to students in both grades (i.e., the wording of item stems was confusing and/or the response options were misleading). However, math participants did invoke two explanations that had not been frequently mentioned by reading participants: content that was taught only in a lower grade and not the upper one, and ceiling effects. The content-based explanation was that common items with reversals often involve a skill or concepts that was taught in the lower grade but was not revisited instructionally in the upper grade. The ceiling effect explanation was that a common item was too easy. As a result, many small item reversals could be understood as merely the manifestation of a ceiling effect. (Recall, however, that this explanation is contradicted empirically by the fact that there is almost no correlation between lower grade  $p$ -values on actual  $p$ -value growth in math.)

As an example of both these explanations taken together, consider the following item. This item is a rewritten version of an actual item in that it consists of the same mathematical concept but with a different number sequence. This was one of three linking items administered across three grades (2, 3, and 4). (All other linking items were only administered across adjacent grades.)

What is the rule for this pattern?

3, 6, 9, 12

a) Add 3

b) Add 4

c) Multiply by 3

d) Multiply by 4

Across Grades 2 to 3, the  $p$  value for an item similar to the one above grew from 71% to 81%. Yet across Grades 3 to 4, the  $p$ -value decreased from 81% to 79%. So, on one hand, this item was viewed as one for which instructional focus occurs predominantly in Grade 3 rather than in Grade 4. On the other hand, the decrease in Grade 4 was thought to be due to the fact that the  $p$ -value as of Grade 3 was already very high.

There were other examples of items where it seemed clear to participants that knowledge and skills were not being “retained” over time. For example,

The average of Stephanie’s test scores is 80. Which data set could have been her test scores?

a) 90, 70, 80, 90

b) 70, 80, 80, 70

c) 70, 60, 80, 40

d) 70, 80, 90, 80

From Grades 4 to 6,  $p$ -values for an item similar to this one were 58%, 68%, and 64%, respectively. However, though the actual item showed a statistically significant reversal from Grade 5 to 6, participants in this rating group felt it was still an item that belonged on the test, one that was “great for measuring growth.” Math participants frequently made distinctions between items that they thought were generally poor or inappropriate, items that were good but not appropriate for showing growth (e.g., “good for assessment, not good to measure growth” or “not a growth problem”), and items that were appropriate for showing growth (e.g., “shows growth”).

Participants in the sixth- to eighth-grade band group generated some specific explanations for the difference between high growth and reversal items. For high growth items the group noted, “Items that showed growth were items that were emphasized in 7th grade. These are of two types. Material students have been introduced to previously but now go into more depth, and new material (linear problems & graphing)” and that “for the items that showed [high] growth, they reflected the emphasis in 8th grade on algebra (algebraic reasoning).” For reversal items the group noted, “We observed that the scores that decreased were mostly from quant reasoning and geometric reasoning. These topics are covered very little in 8th grade. Students would have studied them more in 6th and 7th grades. Therefore there was not much growth.”

### Investigating Additional Hypotheses

The results from the workshop prompted us to explore the (possibly retrofitted) hypotheses given to explain item reversals in reading. Workshop participants had emphasized that the extent to

which students found reading passages engaging might influence their motivation to give their best effort to answer the associated items correctly. If this were a valid explanation, and if passages vary with respect to how well they engage students, then we might expect to see reversals clustered within passages (presumably those that were the least engaging). In fact, we found that for a number of grade pairings, the percentage of items with  $p$ -value reversals were not uniformly distributed across reading passages. Indeed, for some passages almost every nested common item had a  $p$ -value reversal. This lends some credence to the claim that reversals on the reading tests may have been influenced by choice of reading passage.

We also followed up on the notion that reversals might be caused by “overanalyzing” given a shift in the focus of test items going from recall to inference as students advance through the middle school grades. In fact, for grade pairings 5–6, 7–8 and 9–10 reversals were most likely to occur for common items associated with the standard “Inferences about Content, Ideas, and Author’s Choices.” Of interest, in grade pairings 5–6 and 6–7, there are also a considerable number of reversals for items written for the standard “Retell a Story/Restate Information.”

### Limitations

In hindsight, a weakness in our design of the 2nd day of the workshop was that we did not require each group to come to a consensus decision as to whether a given linking item should be included or excluded from the math or reading vertical scale. As a result, we were unable to quantify the impact that removing such items would have had on grade-to-grade growth interpretation along the lines of the graphic shown in Figure 3. Nor did we provide the individual participants during the Day 1 session (prior to presentation of results on Day 2) with a clear way to indicate whether they thought the item should be rejected outright or whether they thought the item was exemplary for showing growth. The addition of these options would provide additional sources of information about item quality from the perspective of supporting growth inferences and would also make it easier to evaluate the extent to which explanations provided on Day 2 were being retrofitted. Finally, we did not distinguish between the comments and ratings provided by participants who were currently practicing teachers and those who were curriculum specialists. Making such distinctions might be valuable in replications of this growth workshop approach.

### SUMMARY

When experienced teachers and instructional experts are asked to predict student growth at the item level, they tended to predict significantly more growth than is observed empirically. Their implicit ranking of items in math and reading with respect to growth was only weakly correlated to the actual ranking. It can be reasonably argued that the weakness of these correlations is not surprising because (a) teachers have not historically been expected to conceptualize student *growth* between grades within a content area on the whole, let alone item by item, and (b) the empirical growth that teachers were expected to predict, on the whole, has little variability, and this observed variability is itself noisy in that it is influenced by sampling error.

Relative to the full set of candidate-linking items that participants rated blindly, reversal items did not stand out, in the sense that participants did not generally predict these items would show less growth relative to all other linking items. Participants were able to distinguish bottom

quartile growth items from top quartile growth items in math but not in reading. It appears that math participants in the workshop may have found it easier to differentiate the content of each item with respect to the curricular and instructional emphasis of the lower and upper grades. This was also clear from the qualitative discussions that took place on the 2nd day of the workshop.

Qualitative explanations for item reversals differed by content area. In reading, participants emphasized what they viewed as a likely relationship between grade level and student motivation. A widely endorsed perspective was that many grade-to-grade item reversals could be explained by the fact that increasingly unmotivated students were coming into contact with increasingly complex (i.e., “boring”) reading passages. In support of this perspective was the finding that item reversals were in fact clustered in certain reading passages. In math, participants were much more likely to attribute reversals to linking items with material that had not been emphasized instructionally in an upper grade after it had been taught in the lower grade. Finding out that students were not retaining what they had presumably been taught was viewed as important information for a vertical scale to convey.

## DISCUSSION

The design approach being taken to create a vertical scale is not often made transparent. Such transparency requires careful articulation of the growth construct of interest. When states work with test vendors to design their vertical scales, it is clearly important to establish whether a pool of candidate linking items is to be drawn at random from a combined across-grade pool (a “domain definition” for growth), from some subdomain overlap between adjacent grade pairs (the “grade-to-grade definition” for growth), or from a common subdomain that overlaps all grades (“the learning progression” definition for growth). The larger the shift in the knowledge, skills, and abilities being assessed from grade to grade, the less these different design approaches should be expected to lead to similar choices in linking items. In the current investigation, the growth definition intended by the state was never made explicit.

Irrespective of the underlying design approach for a vertical scale, the results of our study indicate that a  $p$ -value reversal in and of itself is not a good reason to remove a linking item. In most cases where participants in our item growth workshop thought an item with a reversal should be excluded, it was not because of the reversal per se but because of a construct irrelevant factor that was thought to be responsible for the reversal, such as a presumed change in student motivation. If a vertical scale consists of many linking items like this, then growth may be underestimated. This raises a deeper problem in the design of a vertical scale. If students are never fully motivated in a field test setting, what are the implications for growth inferences?

For a great number of items in both reading and math, participants found there was no legitimate rationale for excluding reversal items. For such items it appeared that students were not retaining content or skills they were expected to have mastered in prior grades. If this is the case, then the vertical scale will have served the useful purpose of bringing to light the hypothesis that students are not learning as much as teachers—and parents—would want or expect. This represents actionable information that could motivate targeted reflection and instructional improvements. In the present context, although the exclusion of reversal items in reading has a practically significant impact on grade-to-grade growth interpretations (see [Figure 3](#)), it could be argued that this difference is within the range that might be expected if one were able to properly quantify random error due to the specific sample of students responding to

specific samples of items as part of the field test. Unfortunately, it is rare to find statements about the uncertainty in grade-to-grade growth interpretations along a vertical scale.

The domain over which growth is being defined across the grades of any vertical scale is critical. Once this has been decided upon, it makes little sense to differentiate between a linking item and a unique item after they have been field tested. That is, criteria should be established for the inclusion of either item, and once these criteria have been met, the proportion of linking items and unique items should not matter (unless the number of linking items per grade pair drops below some minimum threshold). When a linking item is administered to students in adjacent grades, it becomes part of the domain definition; to subsequently exclude the item is to change the definition of the domain.

On the other hand, when a grade-to-grade definition of growth is being advanced, there is an important conceptual distinction between the use of backward- and forward-linking items in that they make it possible to answer fundamentally different counterfactual questions about students in upper and lower grades. The backward-linking design makes it possible to evaluate how upper grade students would perform on lower grade content; the forward-linking design makes it possible to evaluate how lower grade students would perform on upper grade content. If a vertical scale is calibrated after effectively combining the results from both backward- and forward-linking items to estimate linking constants, the question the vertical scale is making it possible to answer with respect to growth is equivocal. This became clear to us when we were forced to choose a verbal description that would help workshop participants understand how the state was conceptualizing grade-to-grade growth. The scenario we provided participants was consistent with a vertical scale calibrated solely with forward-linking items. In general, vertical scales calibrated solely with forward-linking items appear more likely to produce growth across grades that is larger in magnitude than what would be evident from the sole use of backward-linking items. This is because in the forward-linking design, students in the lower grade are essentially being given a pretest consisting of items with content that may be relatively unfamiliar to them, especially in mathematics.

The sorts of investigations that can be facilitated through the item growth workshop we have described here could be improved upon and used whenever there is an interest in placing across grade tests onto a vertical scale. One can imagine this happening either as part of a pilot test before coming to a consensus on a meaningful growth definition for the vertical scale or as part of a validation study after linking items have already been administered in a field test. As the emphasis on assessments designed for the purposes of supporting inferences about growth continues to increase, it will become all the more important that the design stage of creating a vertical scale be placed on equal footing with the calibration stage.

## REFERENCES

- Birnbaum, A. (1968). Some latent trait models and their use in inferring an examinee's ability. In F. M. Lord & M. R. Novick (Eds.), *Statistical theories of mental test scores*. Reading, MA: Addison-Wesley.
- Briggs, D. C. (2011). *Making inferences about growth and value-added: Design issues for the PARCC consortium*. White paper Commissioned by the PARCC Large-Scale Assessment Consortium, 397–479.
- Briggs, D. C. (2013). Measuring growth with vertical scales. *Journal of Educational Measurement*, 50, 204–226.
- Briggs, D. C., & Weeks, J. P. (2009). The impact of vertical scaling decisions on growth interpretations. *Educational Measurement: Issues and Practice*, 28(4), 3–14.



- Camilli, G. (1988). Scale shrinkage and the estimation of latent distribution parameters. *Journal of Educational Statistics*, 13, 227–241.
- Camilli, G. (1999). Measurement error, multidimensionality and scale shrinkage: A reply to Yen and Burkett. *Journal of Educational Measurement*, 36, 73–78.
- Camilli, G., Yamamoto, K., & Wang, M. (1993). Scale shrinkage in vertical equating. *Applied Psychological Measurement*, 17, 379–388.
- Dadey, N. L., & Briggs, D. C. (2012). A meta-analysis of growth trends from vertically scaled assessments. *Practical Assessment, Research & Evaluation*, 17(14). Available from <http://pareonline.net/>
- Kolen, M. J. (2006). Scaling and norming. In R. L. Brennan (Ed.), *Educational measurement* (4th ed.) (pp. 155–186). Westport, CT: American Council on Education and Praeger.
- Kolen, M. J., & Brennan, R. L. (2004). *Test equating, scaling, and linking: Methods and practices* (2nd ed.). New York, NY: Springer.
- Lord, F. M., & Novick, M. R. (1968). *Statistical theories of mental test scores*. Reading, MA: Addison-Wesley.
- Polikoff, M. S. (2010). Instructional sensitivity as a psychometric property of assessments. *Educational Measurement: Issues and Practice*, 29(4), 3–14.
- Popham, J. W. (2007). Instructional insensitivity of tests: Accountability's dire drawback. *Phi Delta Kappan*, 89, 146–155.
- Rasch, G. (1960). *Probabilistic models for some intelligence and attainment tests*. Copenhagen, Denmark: Danish Institute for Educational Research.
- Skaggs, G., & Lissitz, R. W. (1986). IRT test equating: Relevant issues and a review of recent research. *Review of Educational Research*, 56, 495–529.
- Stocking, M. L., & Lord, F. M. (1983). Developing a common metric in item response theory. *Applied Psychological Measurement*, 7, 201–210.
- Young, M. J. (2006). Vertical scales. In S. M. Downing & T. M. Haladyna (Eds.), *Handbook of test development* (pp. 469–485). Mahwah, NJ: Erlbaum.
- Yen, W. M. (1985). Increasing item complexity: A possible cause of scale shrinkage for unidimensional item response theory. *Psychometrika*, 50, 399–410.
- Yen, W. M. (1986). The choice of scale for educational measurement: An IRT perspective. *Journal of Educational Measurement*, 23, 299–325.
- Yen, W. M., & Burket, G. R. (1997). Comparison of item response theory and Thurstone methods of vertical scaling. *Journal of Educational Measurement*, 34, 293–313.